FINAL PROJECT· GROUP 3

2015 — 2024

A DESCRIPTIVE ANALYSIS

Inflation, *interest rates*, and investment performance.

Understanding how changing inflation environments influence market and sector returns.

DATA SOURCES

FRED API · Yahoo Finance

PERIOD

Jan 2015 — Dec 2024 (monthly)

METHODS

EDA, OLS & logistic regression

Research questions.

Inflation affects household budgets every day, but its impact on investment portfolios is less obvious.

Q1 How has inflation changed over time during the 2015–2024 period?

Q2 How did broad market ETFs perform during this period?

Q3 Do certain sectors appear to perform differently during higher vs. lower inflation periods?

Q4 Are there visible patterns between inflation, interest rates, and market returns?

Q5 Do inflation and interest rates help explain broad market returns in a simple regression model?

Data sources.

Four datasets, each pulled programmatically and reconciled to a monthly cadence from January 2015 through December 2024.

DATASET	IDENTIFIER	SOURCE	COVERAGE
Consumer Price Index Headline CPI, all urban consumers	**CPIAUCSL**	FRED API	120 monthly observations
Federal Funds Rate Effective monthly rate	**FEDFUNDS**	FRED API	120 monthly observations
Broad market ETFs SPY, QQQ, DIA — month-end close	**3 tickers**	Yahoo Finance	120 months × 3 series
Sector ETFs SPDR Select Sector series	**9 tickers**	Yahoo Finance	120 months × 9 series

Together, these datasets represent the economic environment and investment choices commonly available to U.S. investors.

All four datasets passed completeness, type, and duplicate checks. No imputations required.

Analytical approach.

A five-step pipeline from raw ingestion to regression — each step gated by data-quality checks.

Step 01

Data collection

CPI and Fed Funds Rate via the FRED API; ETF prices via the Yahoo Finance chart endpoint.

Step 02

Cleaning & validation

Missing values, type integrity, duplicates, and unusual observations were each inspected per series.

Step 03

Feature creation

YoY and MoM inflation, inflation regimes, COVID-period flag, and 12-month ETF returns.

Step 04

Exploratory analysis

Visual comparisons of inflation, rates, and returns across time and macro regimes.

Step 05

Regression

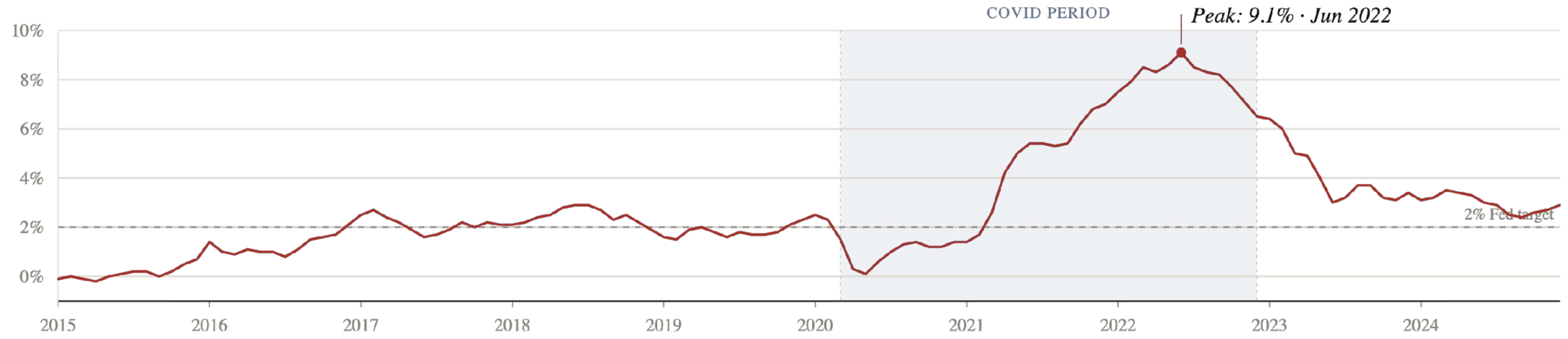
OLS to explain SPY returns; logistic regression to classify high-inflation months.

SQL queries against an in-memory database run alongside each EDA section, mirroring the pandas operations and validating consistency.

Inflation timeline.

CPI year-over-year inflation, 2015 — 2024

Source: FRED CPIAUCSL · monthly



CPI YoY (%)　　COVID period　　Fed 2% target

Inflation reached levels not seen in four decades, affecting housing, groceries, transportation, and investment decisions.

Inflation regimes.

Months are bucketed by CPI YoY into three regimes, providing the basis for cross-environment comparison.

LOW INFLATION

< 2%

55 months

The pre-COVID norm. Predominantly 2015–2020, with average CPI of 1.0% and Fed Funds averaging 0.7%.

MODERATE INFLATION

2 – 4%

39 months

Spans the pre-pandemic late cycle and the post-peak descent. Average Fed Funds: 3.6%.

HIGH INFLATION

> 4%

26 months

Concentrated in 2021–2023. Average CPI: 6.5%. The shock that defines the back half of the sample.

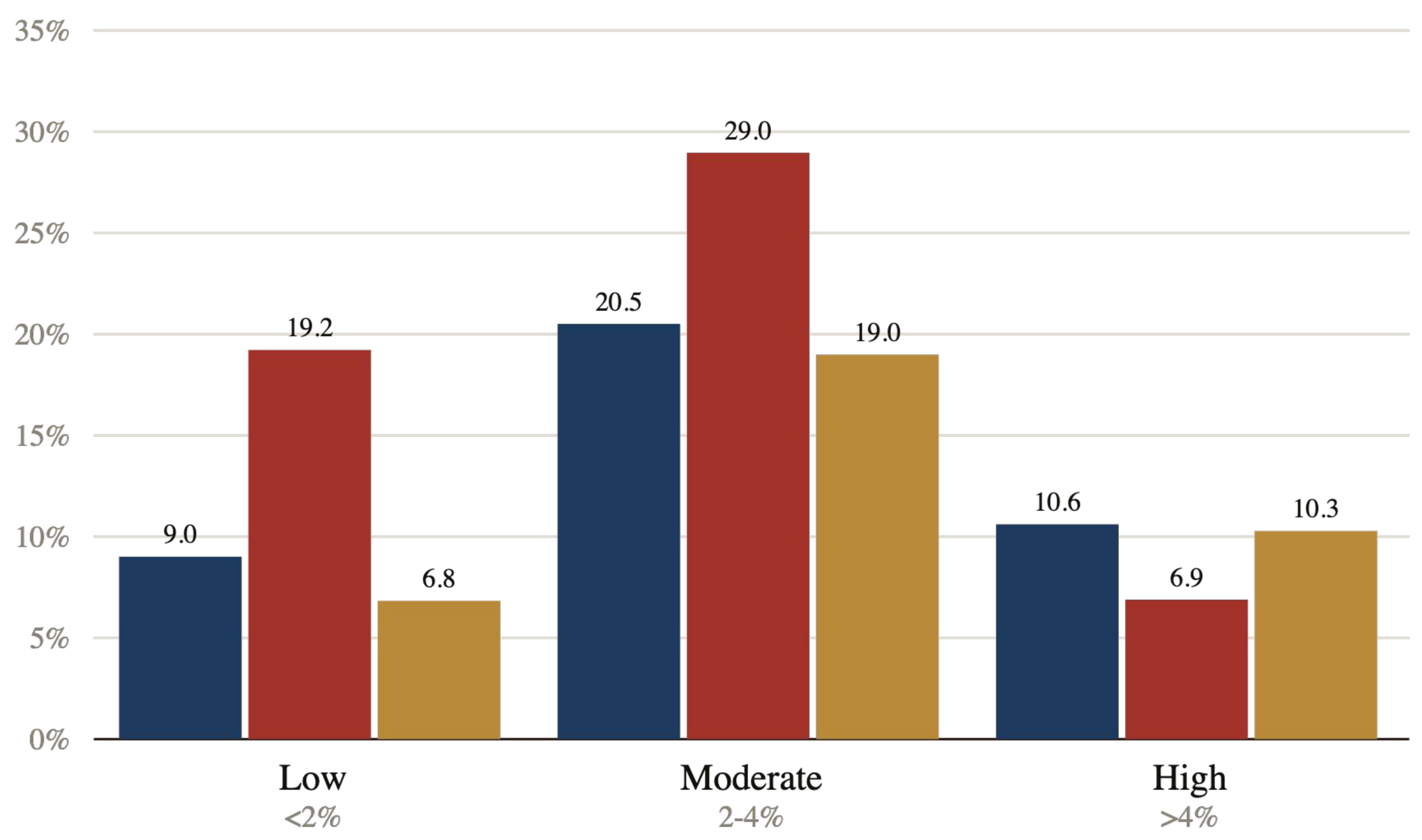
A binary COVID-period flag is also created to separate pandemic-era observations from the rest.

FINDING · Q2

How Did Investors Perform in Different Inflation Environments?

Average 12-month return by inflation regime

SPY · QQQ · DIA



SPY QQQ DIA

PATTERN

Moderate inflation produced the strongest returns across all three broad ETFs.

OUTLIER

QQQ swings the widest — from 28.9% in moderate inflation to 6.9% in high inflation.

STABILITY

SPY and DIA are more regime-stable than QQQ, reflecting broader sector composition.

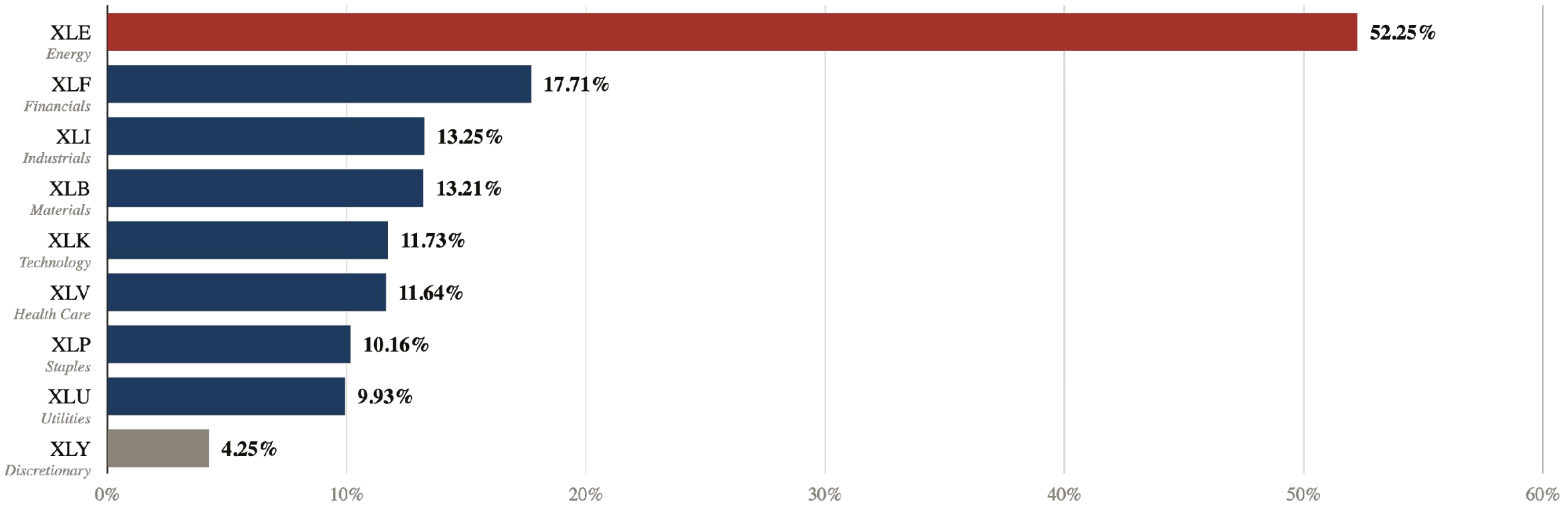
SIGNIFICANCE

Understanding which market environments historically supported stronger returns can help investors contextualize future inflationary periods.

Inflation Creates Winners and Losers

Average 12-month sector ETF return — months with CPI YoY > 4%

Ranked, 9 SPDR sectors



During high inflation, energy stocks dramatically outperformed while consumer-focused companies lagged, reflecting how rising prices affect different parts of the economy.

FINDING · Q5

Can Inflation Predict Market Returns? OLS regression results.

Unsurprisingly, the answer is **largely no.**

A multiple linear regression on 108 monthly observations.

MODEL — OLS · Y = SPY 12-MONTH RETURN

VARIABLE	COEF.	STD. ERR	T	P>|T|
const	17.460	2.683	6.51	**0.000**
CPI_YoY	−0.824	0.625	−1.32	0.191
FedFundsRate	−0.276	0.728	−0.38	0.706

R-SQUARED

0.019

~2% of variance explained

F-STATISTIC

1.01

p = 0.367

INTERPRETATION

Both coefficients carry the expected negative sign, but neither reaches statistical significance. Inflation and rates alone explain very little of broad market return variation. This suggests investors should be cautious about making broad market decisions based solely on inflation expectations.

Logistic regression results.

Flipping the question: can market behavior identify high-inflation months? Predict `HighInflation = 1` from Fed Funds Rate plus SPY, XLE, and XLK returns.

OVERALL ACCURACY

94 %

108 monthly observations

Market signals — particularly XLE returns and the level of the Fed Funds Rate — can fingerprint high-inflation regimes with high accuracy. While inflation may not directly predict market returns, market behavior appears to reflect broader inflationary conditions.

CLASSIFICATION REPORT

CLASS	PRECISION	RECALL	F1	SUPPORT
0 — Not high inflation	0.95	0.98	0.96	82
1 — High inflation	0.92	0.85	0.88	26
Weighted average	**0.94**	**0.94**	**0.94**	**108**

Key findings.

Finding 01

Inflation shifted regimes — twice.

Stable near 2% through 2019, briefly suppressed in 2020, then a sharp spike peaking at ~9% in mid-2022 before partial normalization through 2024.

Finding 02

Rates followed, with a lag.

The Fed Funds Rate moved with inflation but trailed it — months where inflation was already elevated while rates remained below 2% are clearly visible.

Finding 03

Moderate inflation, strongest returns.

Broad ETFs delivered their best average returns in the 2–4% inflation band. Both low and high regimes underperformed it.

Finding 04

Sectors rotated, not aligned.

XLE averaged 52% under high inflation while XLY managed only 4%. Inflation is not a sector-neutral event.

Finding 05

Weak direct linear relationship.

Correlation between CPI and SPY returns is just −0.13. OLS explains ~2% of variance. Inflation alone is not a return predictor.

Finding 06

But the signal runs the other way.

Logistic regression classifies high-inflation months with 94% accuracy from market signals — the macro environment leaves a fingerprint on returns.

Key takeaway

While inflation does not fully explain market returns, it affects the economic environment in which investment decisions are made. Understanding inflation and interest rate trends can help investors better interpret market behavior, evaluate risk, and identify sectors that may perform differently across economic conditions.

Over the long run, investors must earn returns that exceed inflation to preserve purchasing power and maintain the real value of their wealth. Therefore, understanding inflation and interest rate dynamics remains relevant for all investors, from individuals saving for retirement to institutions managing billions of dollars on behalf of clients.

Thank *you.*

Questions and discussion.

Sources: FRED · Yahoo Finance

Methods: pandas · SQL · statsmodels · scikit-learn

Period: Jan 2015 — Dec 2024

Appendix

ETF universe.

Three broad-market ETFs anchor the overall view; nine SPDR Select Sector ETFs allow sector-level comparison.

BROAD MARKET — 3 ETFS

SPY	**S&P 500.** 500 large-cap U.S. companies. The most-traded ETF in the world.
QQQ	**Nasdaq-100.** Tech-heavy — Apple, Microsoft, Nvidia, Meta.
DIA	**Dow Jones Industrial Average.** 30 blue-chip names.

SPDR SELECT SECTOR — 9 ETFS

XLE	**Energy.** ExxonMobil, Chevron.
XLF	**Financials.** JPMorgan, Berkshire Hathaway, Visa.
XLK	**Technology.** Apple, Microsoft, Nvidia.
XLU	**Utilities.** NextEra Energy, Duke Energy.
XLP	**Consumer Staples.** Procter & Gamble, Coca-Cola, Walmart.
XLY	**Consumer Discretionary.** Amazon, Tesla, Home Depot.
XLB	**Materials.** Linde, Sherwin-Williams, Air Products.
XLI	**Industrials.** GE Aerospace, Caterpillar, Union Pacific.
XLV	**Health Care.** UnitedHealth, J&J, Eli Lilly.

FINDING · Q1

Federal Funds Rate timeline.

Effective Federal Funds Rate, 2015 — 2024

Source: FRED FEDFUNDS · monthly

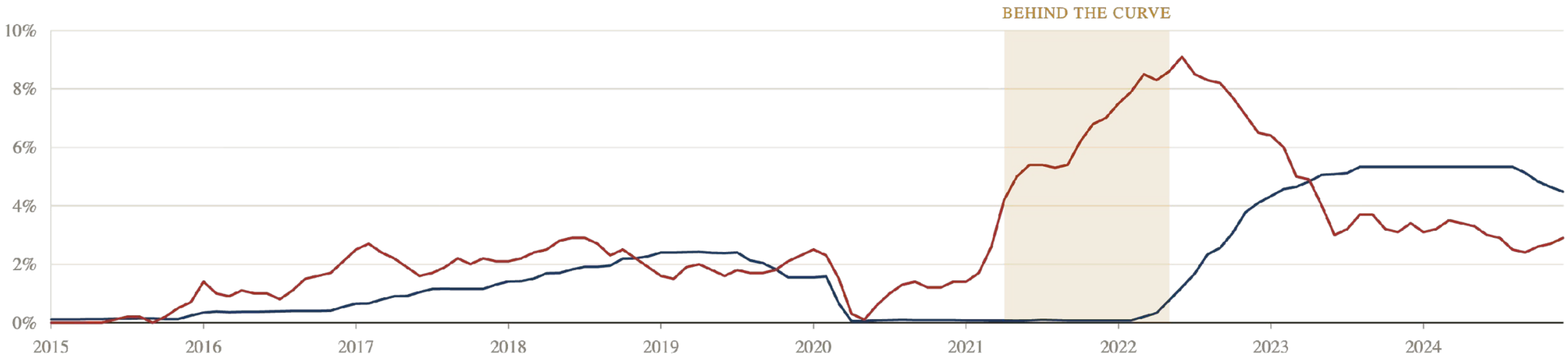
Effective Fed Funds Rate (%)

Lift-off from zero in 2015–2018, a return to the zero lower bound during COVID, then the fastest tightening cycle in four decades — from 0.08% in March 2022 to 5.33% by August 2023.

Inflation and monetary policy.

CPI YoY vs. Federal Funds Rate

Both series, monthly · same axis



CPI YoY (%) Fed Funds Rate (%) Behind-the-curve months

Rates followed inflation rather than leading it — CPI was running above 5% for roughly a year before the Fed began tightening in earnest.

Sector winners and laggards.

Average 12-month returns by regime expose the rotation: energy & financials lead under high inflation; tech & discretionary lead under moderate or low.

SECTOR ETF	LOW (< 2%)	MODERATE (2–4%)	HIGH (> 4%)	SPREAD
XLE — Energy	-15.6%	9.3%	**52.3%**	67.8
XLF — Financials	0.8%	21.0%	**17.7%**	20.2
XLI — Industrials	5.2%	19.4%	13.3%	14.2
XLB — Materials	5.4%	14.2%	13.2%	8.8
XLK — Technology	**20.8%**	**31.2%**	11.7%	19.5
XLV — Health Care	7.6%	12.0%	11.6%	4.4
XLP — Staples	9.2%	7.0%	10.2%	3.2
XLU — Utilities	10.5%	8.9%	9.9%	1.6
XLY — Discretionary	11.4%	19.9%	4.3%	15.6

Correlation overview.

Pairwise correlation: macro variables & ETF 12-month returns

Pearson r

	CPI	FedFunds	SPY	QQQ	DIA	XLE	XLF	XLK	XLU	XLP	XLY	XLB	XLI	XLV
CPI	1.00	0.62	-0.13	-0.21	-0.05	0.71	0.18	-0.28	0.08	0.09	-0.31	0.05	0.06	-0.02
FedFunds	0.62	1.00	-0.08	-0.18	-0.02	0.42	0.22	-0.20	0.05	0.04	-0.22	0.10	0.08	0.03
SPY	-0.13	-0.08	1.00	0.92	0.94	0.45	0.84	0.92	0.62	0.72	0.86	0.83	0.89	0.78
QQQ	-0.21	-0.18	0.92	1.00	0.81	0.30	0.75	0.97	0.52	0.62	0.86	0.74	0.80	0.74
DIA	-0.05	-0.02	0.94	0.81	1.00	0.55	0.87	0.81	0.66	0.74	0.78	0.84	0.91	0.75
XLE	0.71	0.42	0.45	0.30	0.55	1.00	0.58	0.22	0.32	0.30	0.18	0.51	0.49	0.30
XLF	0.18	0.22	0.84	0.75	0.87	0.58	1.00	0.72	0.48	0.55	0.67	0.79	0.85	0.62
XLK	-0.28	-0.20	0.92	0.97	0.81	0.22	0.72	1.00	0.50	0.58	0.85	0.71	0.78	0.72
XLU	0.08	0.05	0.62	0.52	0.66	0.32	0.48	0.50	1.00	0.68	0.46	0.55	0.61	0.55
XLP	0.09	0.04	0.72	0.62	0.74	0.30	0.55	0.58	0.68	1.00	0.58	0.62	0.67	0.65
XLY	-0.31	-0.22	0.86	0.86	0.78	0.18	0.67	0.85	0.46	0.58	1.00	0.72	0.77	0.68
XLB	0.05	0.10	0.83	0.74	0.84	0.51	0.79	0.71	0.55	0.62	0.72	1.00	0.84	0.66
XLI	0.06	0.08	0.89	0.80	0.91	0.49	0.85	0.78	0.61	0.67	0.77	0.84	1.00	0.70
XLV	-0.02	0.03	0.78	0.74	0.75	0.30	0.62	0.72	0.55	0.65	0.68	0.66	0.70	1.00

−1.0 0 +1.0

AS EXPECTED

CPI and Fed Funds are positively correlated — rates follow inflation.

SURPRISE

Inflation shows weak linear correlation with broad ETF returns.

STANDOUTS

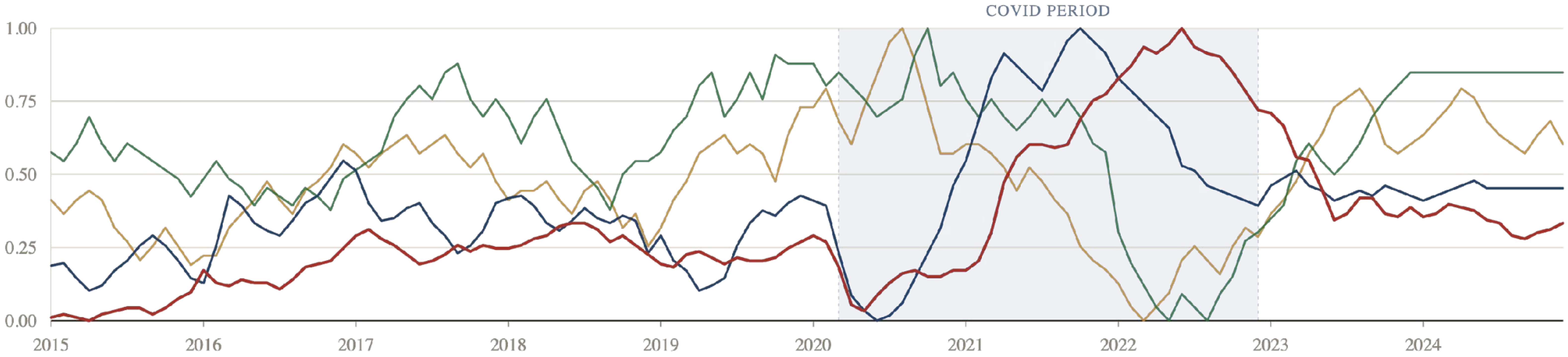
XLE is the only ETF with a strong positive correlation to inflation; XLK and XLY are mildly negative.

FINDING · Q4

Normalized trends.

CPI YoY and ETF 12-month returns, rescaled 0 – 1

Min-max normalized · monthly

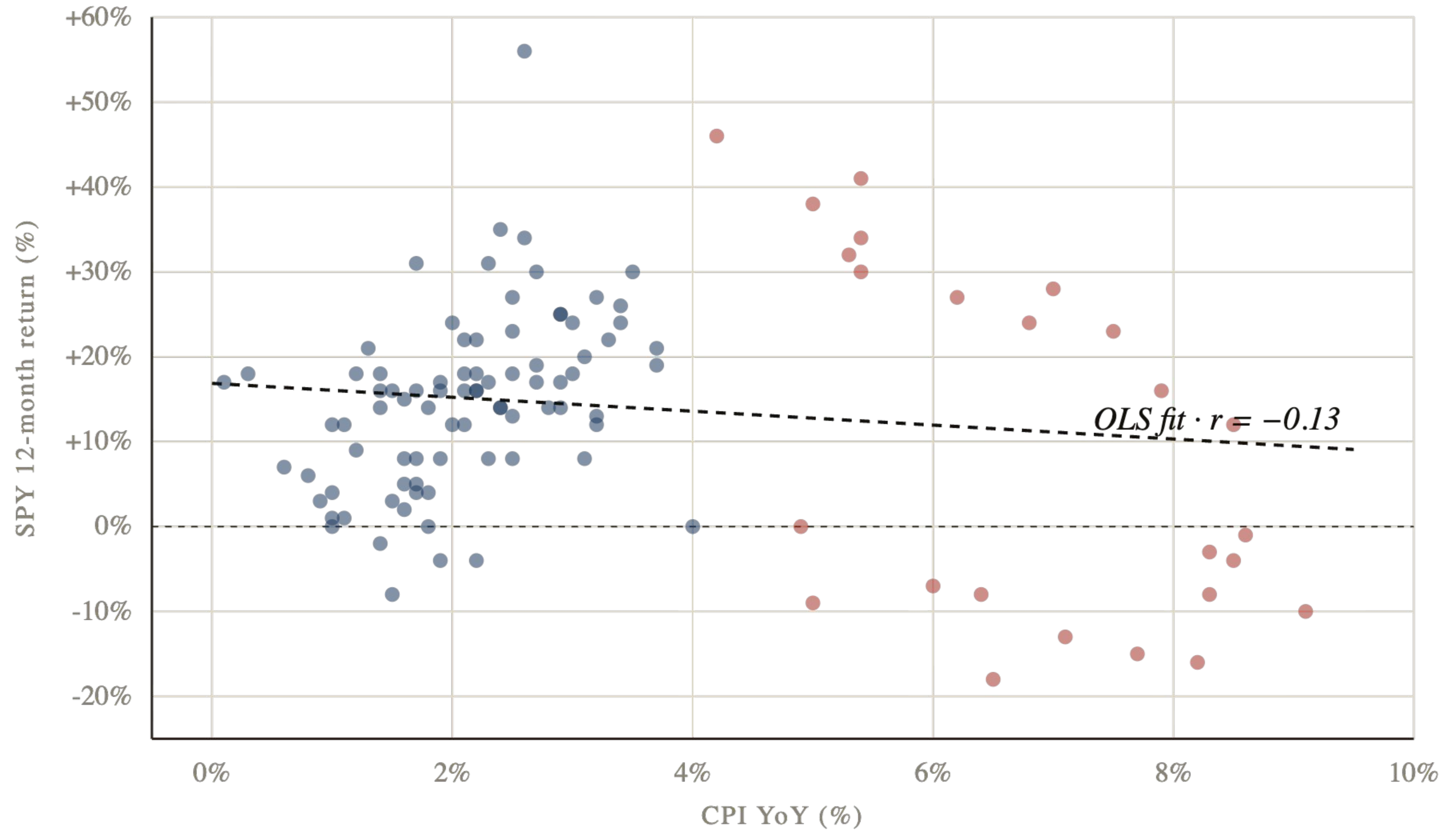


CPI YoY XLE (Energy) XLK (Tech) XLY (Discretionary)

XLE tracks CPI closely; XLK and XLY trend inversely. The same inflation environment is a tailwind for energy and a headwind for tech and consumer spending.

Inflation vs. SPY returns.

CPI YoY (%) vs. SPY 12-month return (%) 108 monthly observations



CORRELATION COEFFICIENT

−0.13 r

Weak, negative, not statistically significant

READING

Higher inflation tends to coincide with weaker SPY performance — but the relationship is loose and inflation alone does not predict returns.

DISPERSION

Return variance widens as inflation rises, suggesting the macro regime affects risk as much as it affects return.

REGIME COMPARISON

COVID vs. non-COVID periods.

Pandemic-era months show a categorically different macro environment — and a partial rotation in which broad ETF held up best.

METRIC	NON-COVID	COVID	Δ
CPI YoY (%) average inflation	2.12	**4.52**	+2.40
Fed Funds Rate (%) average effective rate	2.11	1.79	−0.32
SPY return (%) avg 12-month	15.09	13.30	−1.79
QQQ return (%) avg 12-month	20.02	**20.88**	+0.85
DIA return (%) avg 12-month	15.30	9.76	−5.54

Inflation more than doubled during COVID. Growth-tilted QQQ held up; old-economy DIA absorbed the drag.

CAVEATS

Limitations.

The analysis is descriptive, not causal — four constraints to keep in mind when reading the findings.

LIMITATION	WHAT IT MEANS
01 — Historical	The analysis describes 2015–2024 — a single decade containing one major shock. Patterns observed need not generalize forward.
02 — Narrow inputs	Only CPI, Fed Funds Rate, and ETF returns are included. Earnings, unemployment, GDP growth, and volatility are absent.
03 — Simple model	OLS with two predictors and an in-sample logistic model intentionally trade sophistication for interpretability.
04 — Monthly cadence	Monthly aggregation can mask short-term reactions to specific inflation releases or rate decisions.

FORWARD LOOK

Next steps.

Four directions a follow-on analysis could pursue — each addresses a specific limitation above.

Next 01

Broader macro inputs

Incorporate unemployment, GDP growth, and market-volatility indices to widen the explanatory base.

Next 02

Rolling correlations

Compute rolling 12- and 24-month windows to detect when relationships strengthen or break down.

Next 03

Higher-frequency response

Event-study analysis around CPI releases and FOMC announcements at daily resolution.

Next 04

Advanced modeling

Out-of-sample validation, regularized regression, and tree-based models to quantify non-linear relationships.